

Chapter 4: Consumer's Behavior

Concept of Utility

- **Utility** means satisfaction a consumer gets from consuming goods or services.
 - It measures the **want-satisfying power** of a commodity.
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Approaches of Utility Analysis

- **Cardinal Utility Analysis** – Utility can be measured numerically (in “utils”).
 - **Ordinal Utility Analysis** – Utility can only be ranked, not measured.
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Cardinal Utility Analysis

Assumptions:

1. **Rational consumer** – aims for maximum satisfaction.
 2. **Cardinal measurement** – utility can be measured in numbers.
 3. **Constant marginal utility of money.**
 4. **Diminishing marginal utility.**
 5. **Independent utilities** – utility from one good doesn't affect another.
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Total Utility (TU)

- Total satisfaction from consuming all units.

- **Formula:**

$$TU = MU_1 + MU_2 + \dots + MU_n = \sum MU$$

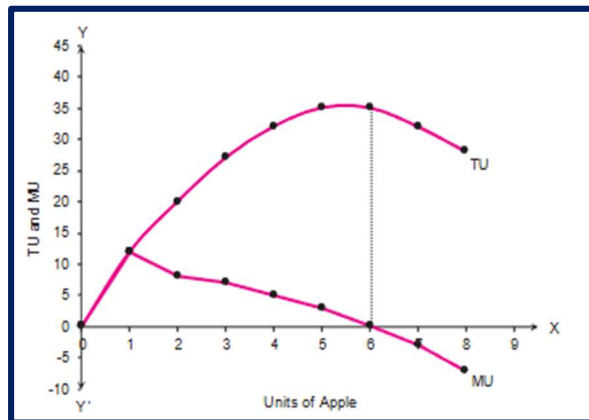
Marginal Utility (MU)

- Extra satisfaction from one more unit.

- **Formula:**

$$MU = \Delta TU / \Delta Q \text{ OR } MU_n = TU_n - TU_{n-1}$$

Relationship between TU and MU



Condition	TU	MU
Increasing satisfaction	↑	+
Maximum satisfaction	Max & constant	0
Decreasing satisfaction	↓	-

Law of Diminishing Marginal Utility

- As more units are consumed, MU decreases.



Assumptions:

Rational consumer, measurable utility, constant MU of money, homogeneous goods, continuous consumption, suitable unit size, unchanged taste/habit.

Exceptions:

Non-homogeneous goods, taste change, breaks in consumption, large units, luxury items, changing income, irrational behavior.

Importance:

Basis of economic laws, taxation, price, consumer spending, and wealth distribution.

Law of Substitution (Equi-Marginal Utility / Gossen's Second Law)

- Consumer gets maximum satisfaction when:

$$MU_x/P_x = MU_y/P_y = MU_m$$

Assumptions:

Cardinal utility, rational consumer, constant income, constant MU of money and prices, divisible goods, fixed time.

Limitations:

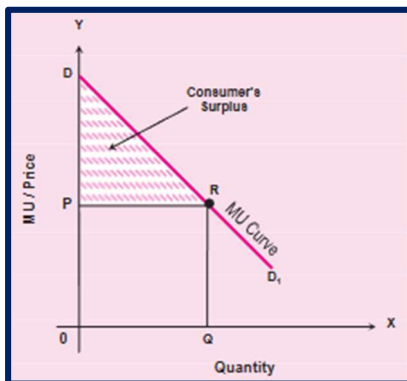
Ignorance, indivisible goods, immeasurable utility, customs, unlimited income, uncertain choice.

Importance:

Used in consumption, production, exchange, distribution, public finance, and price fixing.

Consumer Surplus

- Extra benefit consumer gets when willing to pay more than actual price.



- Formula:**

$$\text{Consumer Surplus} = \text{Total Utility} - \text{Total Expenditure}$$

Assumptions:

Rational consumer, measurable utility, constant MU of money, expected price > actual price, homogeneous units, constant price.

Criticisms:

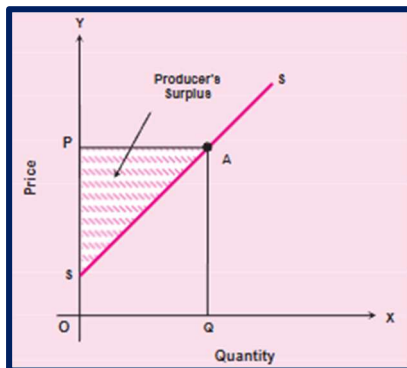
Imaginary, utility not measurable, MU of money not constant, not for necessities, ignores complementary/substitute goods.

Importance:

Used in public finance, business pricing, trade benefits, and value comparisons.

Producer's Surplus

- Extra benefit to producer when selling price exceeds cost.



- **Formula:**

Producer Surplus = Amount Received – Cost of Seller